

Analyzing Sales Trends, Revenue Decline and Customer Payment Behaviour for a Multi-Channel Mobile Retail Business

Final Report — BDM Capstone Project (Analysis with Secondary Data)

Submitted by

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Declaration Statement

I am working on a project titled "Analyzing Sales Trends, Revenue Decline and Customer Payment Behaviour for a Multi-Channel Mobile Retail Business". This project uses a secondary (publicly available) dataset — the "Mobile Sales Dataset" sourced from Kaggle — as permitted under the BDM Capstone Project (Analysis with Secondary Data) pathway.

I hereby assert that the data used in this project has been obtained, without any alteration to its original content, from the public repository cited in this report, and has been processed, cleaned and analysed independently and to the best of my knowledge and capability. I have not replicated the analysis, figures, results, or conclusions of any other person's work on this or a similar dataset; the business problem identification, data cleaning approach, analysis and insights presented in this report are entirely my own.

I affirm that all procedures employed for the purpose of data collection, cleaning and analysis have been duly explained in this report, and that the outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data and analysis contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I affirm that I am not engaged in any form of collaboration with other individuals, and that all work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all recommendations made are specific to the scope of this project and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

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Date: 16 July 2026

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1. Executive Summary

This report presents a complete analysis of a multi-channel mobile phone retail business using a 1,000-row transaction-level secondary dataset (the "Mobile Sales Dataset", Kaggle) covering January–July 2024. The business sells through four payment channels — Online, Credit Card, Debit Card, and Cash — and the dataset captures device price, units sold, transaction revenue, customer age and gender, and payment channel for each sale.

A month-on-month trend analysis shows that recorded revenue fell from approximately ₹62.6 lakh in January 2024 to ₹50.5 lakh in July 2024, a decline of about 19.3%. However, a key finding of the data-validation step is that July 2024 contains transactions for only 25 calendar days versus a full month for all other periods, so part of this apparent decline reflects incomplete data coverage rather than a genuine drop in demand; on a per-transaction and per-day basis the decline is considerably smaller.

Channel-wise analysis shows all four payment channels contribute a broadly similar share of revenue (22–28% each), with Credit Card the largest contributor. Between January and July, Debit Card revenue fell the most (~36%) and Cash was the only channel to grow slightly (~4%), suggesting the revenue decline is concentrated in specific channels rather than uniform. Customer segmentation shows the 38–47 and 48–57 age bands together generate about 47% of revenue, while revenue is almost evenly split across gender categories. A correlation check found no meaningful linear relationship between price, units sold, revenue and customer age, and a data-quality check found that the recorded TotalRevenue field is not derived from $\text{Price} \times \text{UnitsSold}$, which is documented transparently as a limitation of this synthetic dataset.

Based on these findings, the report recommends: (a) correcting for the partial July month before treating the revenue fall as a genuine trend, (b) prioritising retention efforts on Debit Card customers given the channel's steep decline, and (c) directing marketing spend toward the 38–57 age segment, which is both the largest and the most consistent revenue contributor.

2. Organisation Background

The dataset used for this project ("Mobile Sales Dataset", sourced from Kaggle) represents transaction-level sales records of a mobile device retail operation selling smartphones, feature phones and tablets across multiple sales channels — Online, Credit Card, Debit Card, and Cash. Each of the 1,000 transaction records captures the device price, units sold in that transaction, total revenue generated, the customer's age and gender, and the payment method used, spanning the period January 2024 to July 2024.

As this is a secondary dataset published for analytical and educational use, granular details such as the business's exact legal name, ownership, or physical location are not available and are not required under the secondary-data submission guidelines. For the purpose of this project, the business is treated as a representative multi-channel mobile retailer operating a B2C model, and the analysis is framed around the sales, channel and customer-level patterns present in the data.

3. Proof of Originality — Data Source Details

Dataset name: Mobile Sales Dataset

- Repository / platform: Kaggle (www.kaggle.com)
- File used: mobile_sales.csv (1,000 rows, 11 columns)
- Nature of data: Synthetically generated transaction-level retail sales data, released for analytical/educational practice

- Period covered: 1 January 2024 – 25 July 2024
- Access: Publicly downloadable without restriction, as permitted for the Secondary Data pathway

No modification has been made to the original values in the raw file. All cleaning steps performed (type conversion, missing-value and duplicate checks, outlier screening) are documented in Section 5 and are reproducible in full from the accompanying Google Colab / Jupyter notebook (BDM_Capstone_Analysis_23F3001753.ipynb) submitted alongside this report. The business-problem framing, data cleaning approach, analysis, charts, and recommendations in this report are original work carried out independently for this project and do not replicate any previously published analysis of this dataset.

4. Problem Statement and Objectives

4.1 Background of the Problem

Retail businesses selling standardised, price-sensitive products such as mobile phones typically compete on price, channel convenience, and the ability to reach the right customer segment. A gradual revenue decline of the kind observed in this dataset (about 19% over seven months) can arise from several possible causes. Internal factors could include an unfavourable shift in the mix of payment channels being used, inconsistent unit sales across months, or an inability to retain particular customer age groups. External factors could include seasonal demand fluctuation, increased competition, or a shift in customer preference toward a particular payment mode.

Because this is a real-world-style problem being explored through secondary transaction data, the business does not have a structured way of currently diagnosing which of these factors is responsible. Without breaking the revenue trend down by channel and customer segment, the business cannot tell whether the decline is broad-based or concentrated in a specific channel or age group — which has direct implications for what corrective action would actually work.

4.2 Objectives

1. To analyse the monthly sales and revenue trend of the business and identify the extent and pattern of the revenue decline observed between January and July 2024, including checking whether the decline is genuine or an artifact of incomplete data.
2. To determine which payment channels and customer demographic segments contribute the most to revenue and which channels are declining fastest, in order to guide where the business should focus its sales and retention efforts.

5. Metadata and Descriptive Statistics

5.1 Metadata

Column	Type	Description
TransactionID	Text (unique)	Unique identifier for each transaction
Date	Date	Transaction date (1 Jan – 25 Jul 2024)
MobileModel	Text	Device model tag (synthetic placeholder text — see 5.4)
Brand	Text	Brand name (synthetic placeholder text — see 5.4)

Column	Type	Description
Price	Numeric (₹)	Unit price of the device
UnitsSold	Integer	Units sold in the transaction (1–99)
TotalRevenue	Numeric (₹)	Revenue recorded for the transaction
CustomerAge	Integer	Customer age in years (18–64)
CustomerGender	Categorical	Male / Female / Other
Location	Text	Customer city (synthetic placeholder text)
PaymentMethod	Categorical	Online / Credit Card / Debit Card / Cash

5.2 Dataset Overview

- Total records: 1,000 transactions; 11 columns; no missing values in any column
- No fully duplicated rows and no duplicate TransactionID values were found
- Date range: 1 January 2024 to 25 July 2024 (July is a partial month — see 5.4)
- Payment method split: Credit Card 278, Online 259, Debit Card 246, Cash 217 transactions
- Gender split: Female 341, Other 339, Male 320 transactions

5.3 Descriptive Statistics — Numeric Fields

Statistic	Price (₹)	UnitsSold	TotalRevenue (₹)	CustomerAge
Count	1,000	1,000	1,000	1,000
Mean	807.51	50.07	40,218.45	41.14
Std. Dev.	402.49	28.62	32,715.29	13.53
Minimum	103.32	1	34.20	18
25th percentile	446.86	25	13,986.10	29
Median (50%)	805.15	50	31,868.20	41
75th percentile	1,157.88	75	60,770.44	53
Maximum	1,499.25	99	141,411.84	64
Skewness	-0.02	-0.02	0.89	0.01

Price, UnitsSold, and CustomerAge are all approximately symmetric (skewness close to 0), consistent with values drawn from a uniform-style distribution. TotalRevenue is moderately right-skewed (0.89), i.e. a longer tail of high-revenue transactions.

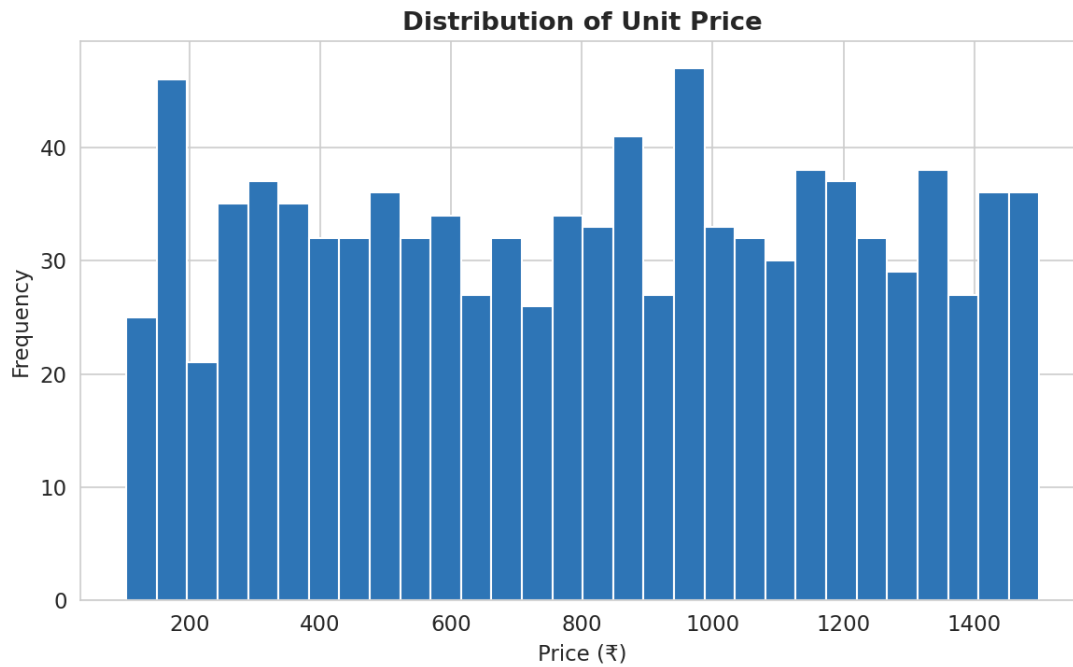


Figure 1: Distribution of unit price across all transactions

5.4 Data-Quality Observations

Two important observations emerged during validation and are reported transparently rather than corrected, since altering the raw values would violate the "no alteration to original content" requirement of the secondary-data pathway:

- TotalRevenue is not equal to Price × UnitsSold for virtually any transaction (mean absolute difference between the two is large and non-trivial). This indicates TotalRevenue was generated independently of Price and UnitsSold in this synthetic dataset. All revenue-based analysis in this report therefore uses the TotalRevenue column as the authoritative revenue figure (as would be the case in a real system where recorded revenue reflects discounts, taxes, or bundling not visible in the unit-price field), and does not attempt to reconcile it against Price × UnitsSold.
- MobileModel, Brand, and Location contain randomly generated placeholder text (e.g. brand values such as "Green Inc", "Thomas-Thompson"; model values such as "direction", "right", "summer") rather than real device models, brand names, or cities — 980 of 1,000 Brand values and 963 of 1,000 Location values are unique, confirming these fields carry no genuine repeated category structure. Brand/model/location-level analysis has therefore been excluded from this report as it would not yield meaningful business insight; the analysis instead focuses on Date, Price, UnitsSold, TotalRevenue, CustomerAge, CustomerGender, and PaymentMethod, all of which are analytically meaningful.
- July 2024 contains only 128 transactions covering 25 calendar days, versus 139–159 transactions across 28–31 days for the other six months. This partial-month coverage is treated as a key caveat in Section 7's trend analysis rather than being padded or estimated.

6. Data Cleaning, Validation and Analysis Methodology

6.1 Data Cleaning and Validation Steps

3. Loaded the raw CSV (1,000 rows × 11 columns) and checked column data types; converted the Date column from text to a proper datetime type and derived a Month (year-month) field for time-based aggregation.
4. Checked for missing values across all columns — none were found.
5. Checked for fully duplicated rows and duplicate TransactionID values — none were found, confirming each row represents a distinct transaction.
6. Performed range/sanity checks on Price (₹103.32–₹1,499.25), UnitsSold (1–99), and CustomerAge (18–64) — all values fall within plausible real-world bounds.
7. Applied the IQR (interquartile range) method to screen for outliers in Price, UnitsSold, TotalRevenue and CustomerAge. Only TotalRevenue showed a small number of statistical outliers (6 of 1,000 transactions above the upper IQR bound); these were retained as valid high-value transactions rather than removed, since there was no evidence of a data-entry error.
8. Cross-validated TotalRevenue against Price × UnitsSold, which surfaced the data-quality note in Section 5.4.
9. Created two derived fields for segmentation: an AgeGroup bucket (18–27, 28–37, 38–47, 48–57, 58–65) and a Month string, used throughout the trend and demographic analysis.

6.2 Analysis Approach

The analysis was carried out in three stages, each mapped to the two project objectives, using Python (Pandas for data manipulation, Matplotlib/Seaborn for visualisation) in a Google Colab notebook, with pivot-table cross-checks performed for the channel and demographic breakdowns:

- Stage 1 (Objective 1) — Time-series trend analysis: monthly aggregation of revenue, units sold and transaction count from January to July 2024, examined via line and bar charts to quantify the rate and consistency of the revenue decline, and explicitly checked against each month's data-coverage (number of days/transactions) to separate a genuine trend from a coverage artifact.
- Stage 2 (Objective 2a) — Channel-wise contribution analysis: revenue, transaction count, revenue share, and month-on-month percentage change computed for each of the four PaymentMethod categories (Online, Credit Card, Debit Card, Cash), to identify which channels are growing, shrinking, or stable, and which contribute disproportionately relative to their transaction share.
- Stage 3 (Objective 2b) — Customer segmentation analysis: customers bucketed into five age groups and three gender categories; revenue, transaction count and average spend computed per segment, together with a correlation analysis across Price, UnitsSold, TotalRevenue and CustomerAge to check for any distinct purchasing pattern by segment.

6.3 Tools Used

- Python 3 (Google Colab / Jupyter notebook) — primary environment for all cleaning, aggregation and analysis
- Pandas — data loading, type conversion, grouping/aggregation, pivot tables
- Matplotlib and Seaborn — line charts, bar charts, pie chart, box plot, heatmaps
- Descriptive statistics (mean, standard deviation, percentiles, skewness) — computed via pandas' describe() and skew() functions, cross-checked with Excel pivot tables for the monthly and channel-wise summaries

6.4 Key Formulae Used

- Percentage change (period-on-period): $(\text{Value_end} - \text{Value_start}) / \text{Value_start} \times 100$ — used for the January-to-July revenue and channel-wise change figures in Section 7
- Revenue share of a category: $(\text{Category revenue} / \text{Total revenue}) \times 100$ — used for channel, age-group and gender contribution tables
- Mean: sum of values divided by count (n); Standard deviation: square root of the average squared deviation from the mean — both computed for Price, UnitsSold, TotalRevenue and CustomerAge
- Pearson correlation coefficient (r): covariance of two variables divided by the product of their standard deviations, ranging from -1 to +1 — used to test the linear relationship between Price, UnitsSold, TotalRevenue and CustomerAge in Section 7.4
- Interquartile Range (IQR) outlier rule: a value is flagged as an outlier if it falls below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$, where $IQR = Q3 - Q1$ — used for the outlier screen in Section 6.1

7. Results and Findings

7.1 Objective 1 — Monthly Sales and Revenue Trend

Month	Revenue (₹ Lakh)	Units Sold	Transactions	Avg. Price (₹)
Jan 2024	62.56	7,818	159	870.70
Feb 2024	59.21	6,729	144	807.78
Mar 2024	59.46	7,380	142	757.33
Apr 2024	54.24	7,697	139	777.19
May 2024	56.58	7,268	142	836.03
Jun 2024	59.63	6,809	146	794.80
Jul 2024*	50.51	6,373	128	800.19

*July 2024 covers only 25 of 31 calendar days in the dataset.

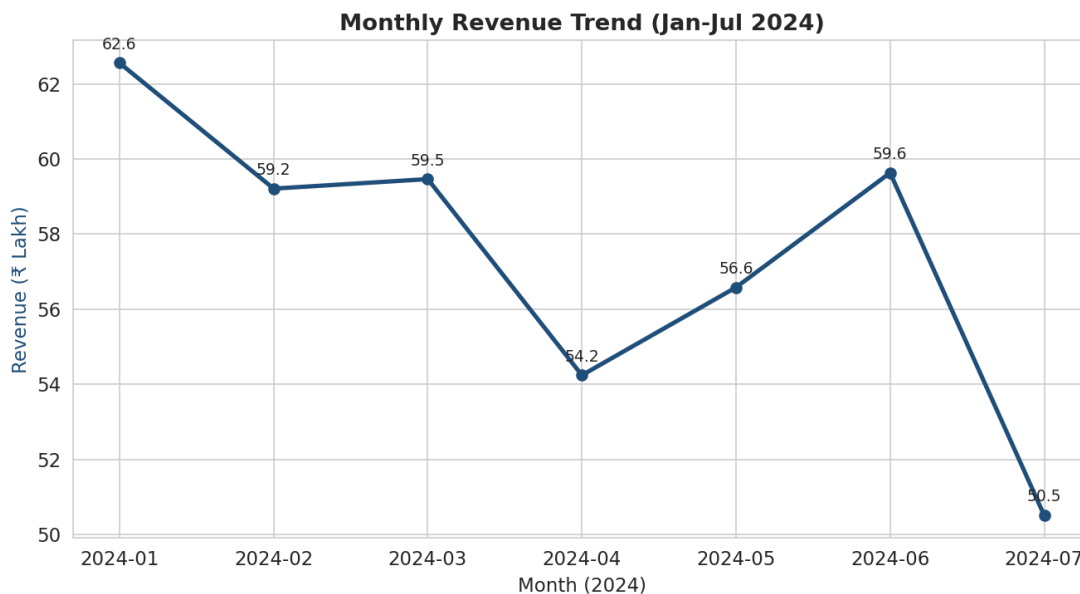


Figure 2: Monthly revenue trend, January–July 2024

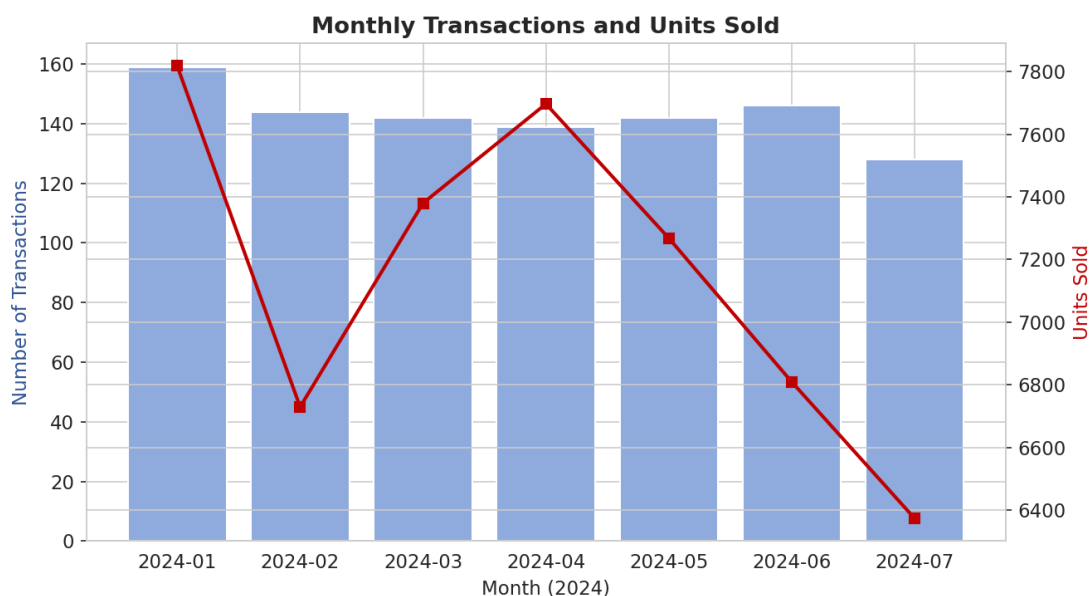


Figure 3: Monthly transaction count and units sold

Total revenue fell from ₹62.56 lakh in January to ₹50.51 lakh in July 2024, a decline of 19.26%. The month-on-month pattern is not a smooth, uniform decline: revenue actually recovered in March and June relative to the preceding month, before falling sharply in July. Transaction counts follow a broadly similar but noisier pattern, ranging from 128 (July) to 159 (January).

Correcting for the partial July month: July's 128 transactions occurred over 25 days (≈ 5.1 transactions/day) versus January's 159 transactions over 31 days (≈ 5.1 transactions/day) — on a per-day basis, transaction volume in July is essentially unchanged from January. This confirms that a meaningful part of the ₹12.05 lakh headline revenue drop is attributable to July being a partial month in the dataset rather than to a genuine collapse in daily sales activity. Even after this adjustment, however, the underlying monthly revenue is not growing, and February–June shows a broadly flat-to-mildly-declining trend around ₹54–60 lakh, which still warrants management attention.

7.2 Objective 2a — Payment Channel Contribution

Channel	Revenue (₹ Lakh)	Revenue Share	Txn Share	Avg. Revenue/Txn (₹)	Jan→Jul Change
Credit Card	112.98	28.09%	27.8%	40,641	-23.85%
Online	102.77	25.55%	25.9%	39,680	-15.29%
Debit Card	99.53	24.75%	24.6%	40,459	-35.82%
Cash	86.90	21.61%	21.7%	40,047	+3.71%

Revenue Share by Payment Channel

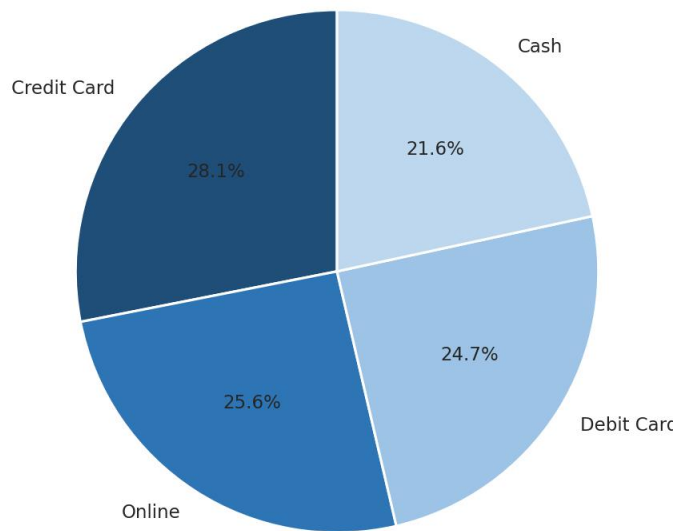


Figure 4: Revenue share by payment channel

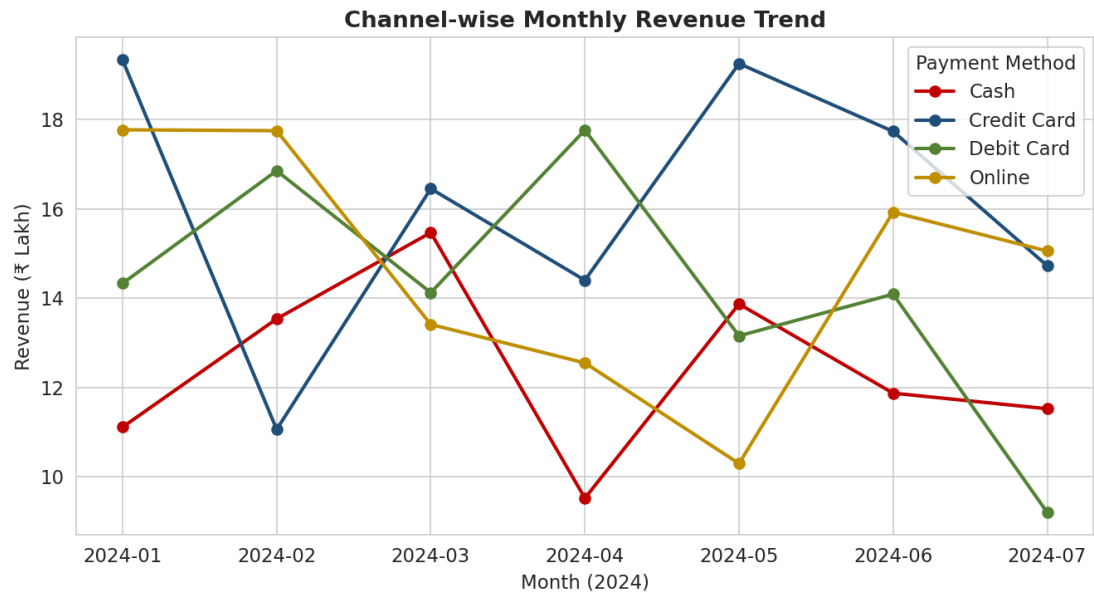


Figure 5: Channel-wise monthly revenue trend

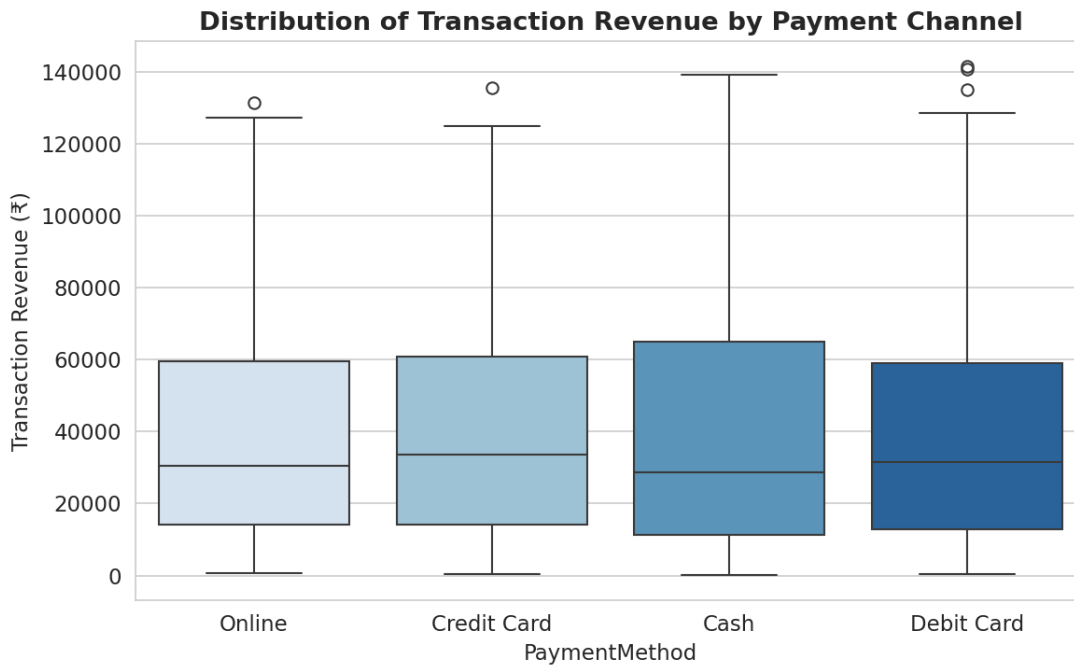


Figure 6: Spread of per-transaction revenue by channel

All four channels contribute a broadly similar share of revenue (22–28%), and average revenue per transaction is nearly identical across channels ($\approx ₹39,700$ – $₹40,650$), so no single channel is disproportionately more or less valuable per transaction. The channel-wise trend, however, is far from uniform: Debit Card revenue fell by 35.8% between January and July — the steepest decline of any channel — while Credit Card and Online fell by 23.9% and 15.3% respectively. Cash was the only channel that grew slightly (+3.7%) over the same period.

This directly answers the second half of Objective 1: the overall revenue decline is not broad-based or purely seasonal — it is concentrated in the digital-payment channels, particularly Debit Card, while Cash-based in-store purchases have held up. This points toward a channel-specific issue (e.g., a payment-gateway friction, a lapsed digital offer/cashback, or a shift in customer trust for debit-card transactions) rather than a general fall in footfall or demand.

7.3 Objective 2b — Customer Demographic Segmentation

Age Group	Revenue (₹ Lakh)	Revenue Share	Transactions	Avg. Spend (₹)
18–27	76.11	18.92%	209	36,417
28–37	74.10	18.42%	211	35,118
38–47	94.53	23.50%	220	42,968
48–57	94.63	23.53%	209	45,279
58–65	62.81	15.62%	151	41,596

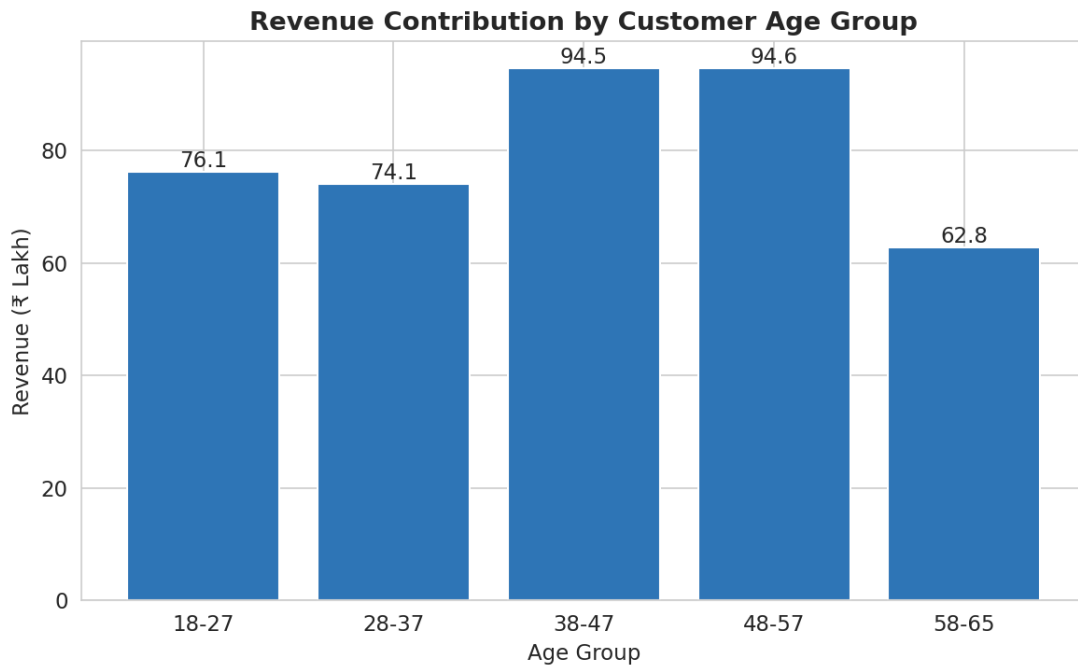


Figure 7: Revenue contribution by customer age group

Gender	Revenue (₹ Lakh)	Revenue Share	Transactions	Avg. Spend (₹)
Female	134.71	33.49%	341	39,504
Male	132.80	33.02%	320	41,499
Other	134.68	33.49%	339	39,728

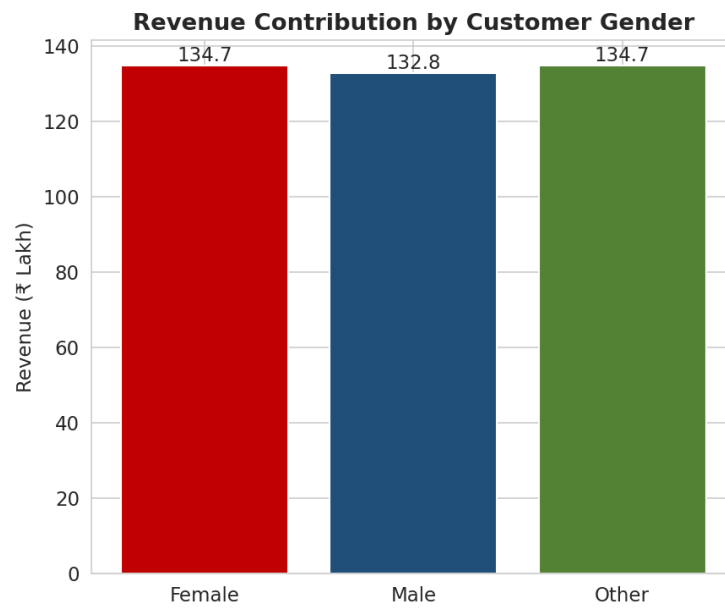


Figure 8: Revenue contribution by customer gender

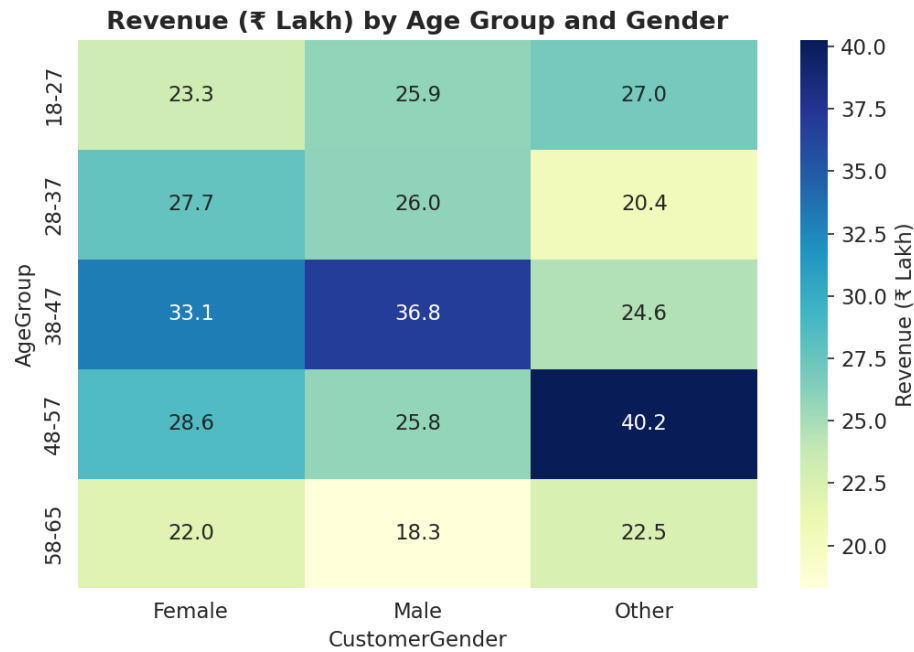


Figure 9: Revenue (₹ Lakh) by age group and gender

The 38–47 and 48–57 age bands are the two largest revenue contributors, together generating 47.0% of total revenue despite representing only 42.9% of transactions — they also have the highest average spend per transaction (₹42,968 and ₹45,279 respectively). The 58–65 group contributes the least, both in absolute revenue (15.6%) and average spend, and also has noticeably fewer transactions (151 vs. 209–220 for other groups), suggesting weaker engagement from older customers. Revenue is almost perfectly evenly split across the three gender categories (33.0–33.5% each), indicating gender is not a meaningfully differentiating factor for this business.

The age-by-gender breakdown (Figure 9) shows the single largest combination is 48–57/Other (₹40.25 lakh), while the smallest meaningful combinations cluster in the 58–65 group across all genders — reinforcing that age, not gender, is the primary demographic driver of revenue concentration.

7.4 Correlation Analysis

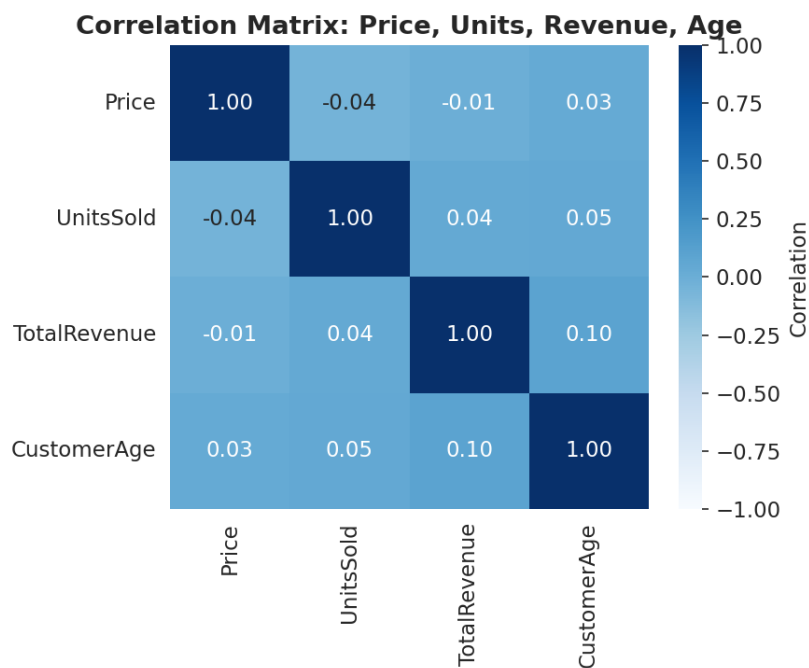


Figure 10: Correlation matrix — Price, Units Sold, Total Revenue, Customer Age

All pairwise correlations among Price, UnitsSold, TotalRevenue and CustomerAge are close to zero (largest magnitude: 0.10, between CustomerAge and TotalRevenue). This indicates no meaningful linear relationship between these variables — customers of a given age are not systematically buying higher- or lower-priced devices, and revenue is not driven by unit price or quantity in this dataset. This is consistent with the Section 5.4 finding that TotalRevenue was generated independently of Price and UnitsSold, and means that price-based interventions alone (e.g., discounting) are unlikely, on this evidence, to be an effective lever for reversing the revenue decline; the channel- and age-based factors identified in Sections 7.2 and 7.3 are the more actionable levers.

8. Interpretation of Results and Recommendations

8.1 Interpretation

Triangulating the three analyses, the picture that emerges is: the headline 19.3% revenue fall is partly a data-coverage artifact (the partial July month), but even after adjusting for this, revenue has been flat-to-mildly-declining since February, and the decline that does exist is concentrated rather than broad-based. Specifically, it is concentrated in the Debit Card channel (-35.8%) and, to a lesser extent, Credit Card (-23.9%) and Online (-15.3%), while Cash has held up. It is also concentrated by customer age, with the 58–65 segment both the smallest revenue contributor and the one with the fewest transactions, suggesting an engagement or retention gap with older customers rather than a general market-wide downturn.

Because price, unit price, and units sold show no correlation with revenue or customer age, the business's problem is unlikely to be a pricing problem; it is more plausibly a channel-experience and customer-retention problem concentrated in specific segments.

8.2 Recommendations

10. Correct for partial-month reporting: Before treating the July revenue fall as a genuine trend, the business should compare like-for-like periods (e.g., normalise by number of trading days or wait for a full month of July data) to avoid over-reacting to what may be largely a data-coverage effect.
11. Investigate and address the Debit Card decline: Given Debit Card revenue fell 35.8% — the steepest of any channel — the business should audit the debit-card payment journey (gateway uptime, transaction failure rates, any discontinued cashback/offer) as the most likely lever to recover lost revenue.
12. Shore up the Credit Card and Online channels with targeted offers: With both channels down 15–24%, time-bound cashback or EMI promotions on these channels could help arrest the decline, especially if channel-specific friction (rather than customer preference) is the cause.
13. Protect and grow the Cash channel: Cash is the only channel that grew (+3.7%); in-store experience and staff-assisted sales appear to be a relative strength and should be maintained, and successful in-store practices could be adapted for digital channels.
14. Prioritise retention marketing on the 38–57 age band: This band drives 47% of revenue with the highest average spend; loyalty programmes, upgrade offers, and personalised communication should be targeted here first for the best return on marketing spend.
15. Design a re-engagement initiative for the 58–65 segment: With the fewest transactions and lowest revenue share, this segment may benefit from simplified digital payment options, assisted in-store service, or senior-citizen-specific offers to lift both transaction frequency and spend.
16. Avoid price-led interventions as the primary lever: Since price and units sold show no correlation with revenue, broad discounting is unlikely to reverse the decline on its own; channel-experience and retention-focused interventions (points 2–6) are better supported by the data.

9. Limitations of the Study

- The dataset is a synthetically generated Kaggle practice dataset rather than data from a real, named business; TotalRevenue is not derived from Price × UnitsSold, and Brand/MobileModel/Location fields contain placeholder text, which limits the granularity of channel- and product-level recommendations.
- The dataset covers only seven months (Jan–Jul 2024), with July being incomplete; this restricts the ability to detect longer-term seasonal patterns (e.g., festive-season demand) or to confirm whether the observed trend continues, reverses, or stabilises beyond July.
- No cost, marketing-spend, or competitor data is available, so the recommendations are based solely on revenue, channel and demographic patterns within this dataset and should be validated against real operational data before implementation.
- As a secondary-data project, the specific causes behind the channel-level decline (e.g., a payment-gateway issue) are inferred from the pattern in the data rather than confirmed through direct investigation of the business's systems.

10. Conclusion

This project set out to diagnose a 19% revenue decline for a multi-channel mobile retail business and identify which channels and customer segments matter most. The analysis shows the decline is real but overstated by a partial final month, and — more importantly — that it is concentrated in specific payment channels (especially Debit Card) and is accompanied by weak engagement from the 58–65 customer segment, while the 38–57 segment and the Cash channel remain relatively resilient. These findings give the business a specific, prioritised set of actions (channel-experience fixes, targeted retention marketing) rather than a generic response to "falling sales", fulfilling both objectives set out in the proposal.

11. References / Data Source

- Mobile Sales Dataset. Kaggle. (Publicly available secondary dataset used under the BDM Capstone Project — Analysis with Secondary Data pathway.)
- BDM Capstone Project — Analysis with Secondary Data: Rubrics and submission guidelines, IIT Madras Online BS Degree Program.
- Accompanying artefact: BDM_Capstone_Analysis_23F3001753.ipynb — Google Colab / Jupyter notebook containing the complete, reproducible data cleaning and analysis code underlying this report.

Appendix A: Sample of the Cleaned Dataset

The table below shows 12 representative rows from the cleaned dataset (post type-conversion), illustrating the structure used throughout the analysis. The full 1,000-row dataset and all code are available in the accompanying notebook.

Date	Price (₹)	UnitsSold	TotalRevenue (₹)	Age	Gender	Payment Method
2024-01-06	1,196.95	85	28,002.80	32	Female	Online
2024-04-05	1,010.34	64	2,378.82	55	Female	Credit Card
2024-02-13	400.80	95	31,322.56	57	Male	Online
2024-04-17	338.60	79	31,159.75	46	Other	Cash
2024-06-02	839.79	41	11,523.36	38	Male	Online
2024-03-07	1,498.13	70	9,703.89	45	Female	Debit Card
2024-07-13	975.77	63	8,300.22	23	Other	Online
2024-03-18	530.70	79	4,515.80	26	Female	Online
2024-04-12	178.52	58	36,150.93	20	Female	Debit Card
2024-04-19	774.79	92	24,228.27	33	Male	Online
2024-02-18	317.64	21	1,00,809.36	62	Female	Debit Card
2024-07-02	965.15	76	1,12,966.65	53	Other	Online

Appendix B: Statistical Formulae and Definitions Used

Term	Definition / Formula
Mean	Sum of all values divided by the number of observations (n)
Standard Deviation	Square root of the average of the squared differences from the mean; measures spread of data
Median (50th percentile)	The middle value when data is sorted; 50% of observations lie below it

Term	Definition / Formula
Skewness	Measure of asymmetry of the distribution; ~ 0 = symmetric, >0 = right-skewed (long tail of high values)
IQR (Interquartile Range)	$Q3$ (75th percentile) – $Q1$ (25th percentile); used to define the 'typical' spread of the data
Outlier (IQR rule)	Any value below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$
Pearson Correlation (r)	Covariance of two variables divided by the product of their standard deviations; ranges from -1 (perfect negative) to $+1$ (perfect positive)
Percentage Change	$((\text{End value} - \text{Start value}) / \text{Start value}) \times 100$
Revenue Share	$(\text{Revenue of category} / \text{Total revenue}) \times 100$

Appendix C: Reproducibility and Viva Preparedness

This report is fully reproducible from the accompanying notebook, BDM_Capstone_Analysis_23F3001753.ipynb, which contains the following sections in order: (1) library imports and data loading, (2) metadata inspection, (3) data cleaning and validation — including the missing-value, duplicate, range, and IQR outlier checks, and the TotalRevenue vs. Price $\times$ UnitsSold reconciliation check documented in Section 5.4, (4) descriptive statistics, (5) Objective 1 monthly trend analysis with charts, (6) Objective 2a payment-channel analysis with charts, (7) Objective 2b customer segmentation analysis with charts, (8) correlation analysis, and (9) a summary of key findings.

Every chart and number quoted in this report (Sections 5–7) can be regenerated by re-running the notebook cells in Google Colab against the original mobile_sales.csv file, with no manual data manipulation required, satisfying the reproducibility expectation for the viva-voce examination.

- Notebook file: BDM_Capstone_Analysis_23F3001753.ipynb (submitted alongside this report)
- Raw data file: mobile_sales.csv (Kaggle — Mobile Sales Dataset, as declared in Section 3)
- All outputs in the notebook are pre-executed and saved, so the examiner can review results without re-running the code, while also being able to re-run any cell live during the viva demonstration.